

# INDIAN INSTITUTE OF TECHNOLOGY JAMMU

## EE-1-01(TO): Basic Electrical and Electronics Engineering End Semester Examination, November 2025

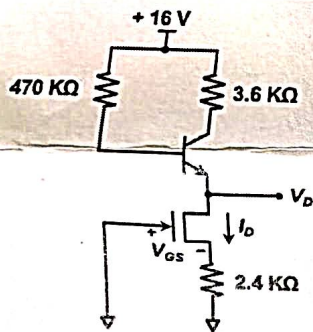
Maximum Marks: 50

Time: 03 Hours

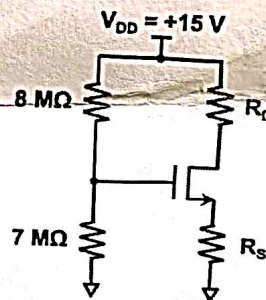
### Important Note:

- Attempt all the questions. Marks are assigned against each question.
- All the answers should be to the point. Negative mark will be awarded for the unnecessary explanations.

- Q1 A 250 V shunt motor on no load runs at 1500 rpm and takes 5A. The total armature and shunt field resistance are 0.2 Ohm and 250 Ohm, respectively. Calculate the speed when loaded and take a current of 50 A if the field weakens by 5%. 04
- Q2 Calculate  $V_D$  for the given circuit-A if the following parameters are given: for BJT  $\beta = 80$ , for JFET  $I_{dss} = 8\text{mA}$  and  $V_p = -4\text{V}$ . 04
- Q3 Consider MOSFET in the circuit-B is operating in the saturation region and the characteristics is given by  $I_D = 0.5(V_{GS} - 1)^2 \text{mA}$ , where  $V_{GS}$  is in V. Find the value of  $R_S$  if  $V_S = +5\text{V}$ . 04

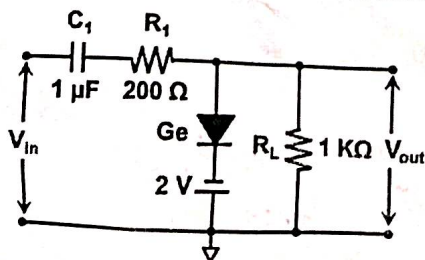


Circuit-A

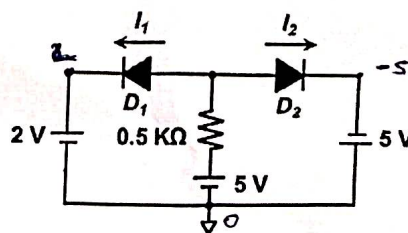


Circuit-B

- Q4 In the given circuit-C, the diode is non-ideal and fabricated from Ge semiconductor. Draw the output waveform and comment on the type of modified circuit if: 05
- (a)  $C_1 = \infty$  and  $V_{in}$  is sine wave with  $V_{av} = 0$  and  $V_m = \pm 10\text{V}$ .
- (b)  $R_1 = 0$  and  $V_{in}$  is square wave with  $V_{P.P} = 30\text{V}$  and  $V_{av} = -5\text{V}$ .
- Q5 In the below circuit-D, consider diodes are Si based. Find the value of  $I_1$  and  $I_2$ . 04



Circuit-C



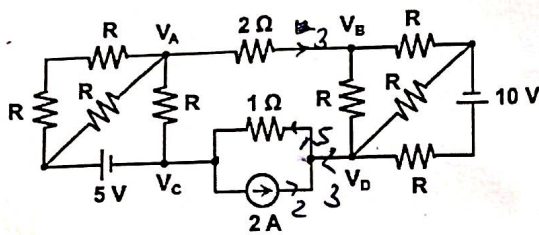
Circuit-D

P.T.O.

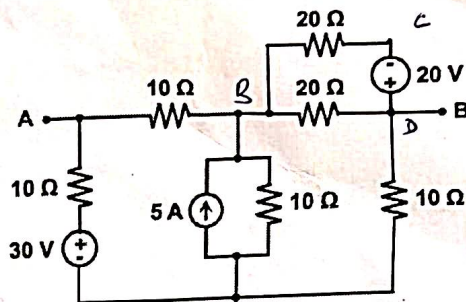
Q6 A single phase 230V, 1 KW heater is connected across single phase 230V, 50Hz supply through a diode. Calculate the power delivered to the heater element. Also find the peak diode current and input power factor. 04

Q7 In the given circuit-E, if  $V_A - V_B = 6V$  then what is the  $V_C - V_D$ ? 04

Q8 Find the Thevenin equivalent across terminal A and B in the circuit-F. 04



Circuit-E

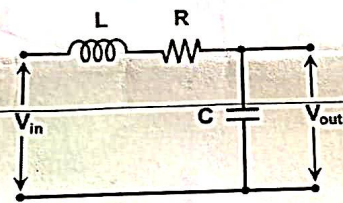


Circuit-F

$$\begin{aligned} \frac{A-B}{10} + \frac{A-30}{10} &= 0 \\ 2A - 30 &= 30 \\ \frac{C-B}{20} + \frac{D-B+0-0}{20} &= 0 \\ \frac{B-A}{10} + \frac{D-C+B-0}{20} &= 0 \\ D-C &= 20 \end{aligned}$$

Q9 A circuit draws 330W from an 110V, 60Hz AC line. The power factor is 0.6 and the current lags the voltage. Find the capacitance value that will result in a power factor of unity. 04

Q10 Sketch the frequency response of the given circuit-G and comment on the change in cut-off frequency for  $L = L_1$  and  $L = L_2$ , considering  $L_1 > L_2$ . 03



Circuit-G

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ \phi &= \tan^{-1}(\omega L - \frac{1}{\omega C}) \\ \omega &= \frac{1}{\sqrt{LC}} \end{aligned}$$

Q11 Briefly explain the following: (Max. 75 words). 02x5

- Can we take one slab of p-type semiconductor and physically join it with an n-type semiconductor to get a p-n junction? Explain.
- What is back EMF? If the back EMF of a DC motor is wiped out suddenly, then what will happen?
- How does temperature affect the operation of a BJT? Which one is preferred for chip manufacturing FET or BJT and why?
- What is the operating condition of the n-channel MOSFET for the following parameters:  $V_{Th} = 2V$ ,  $V_{GS} = 5V$ , and  $V_{DS} = 2V$ ? Neglect the channel length modulation.
- A JFET has the pinch-off voltage of  $-4V$ . What will the drain current when  $V_{GS}$  reached to the 60% of the pinch off voltage?

$$\begin{aligned} I_D &= I_{DSS} (1 - \frac{V_{GS}}{V_{Th}})^2 \\ I_D &= 0.16 I_{DSS} \end{aligned}$$

\*\*\*\*\* ALL THE BEST \*\*\*\*\*